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IT24100391	De Silva T.R.S
IT24100267	De Silva
IT24100070	Anupama M.L.M.
IT24100167	Padmasiri P.D.T.H

Executive Summary

This project develops a data warehouse and business intelligence solution for analysing supply-chain order, sales, customer, product, geographical and delivery information. The DataCo Smart Supply Chain dataset was selected because it contains detailed transactional records covering provisioning, production, sales and commercial distribution activities. The project converts the source data into an analytical architecture consisting of a staging layer, a dimensional data warehouse, a deliveryperformance data mart and a Power BI reporting layer.

The implementation was performed using Microsoft SQL Server and SQL Server Management Studio for database creation, data profiling, transformation, loading and validation. A dimensional model was designed around an order-item fact table supported by customer, product, date, location and shipping dimensions. The order-item grain was selected because it preserves the most detailed transactional level required for quantity, sales, discount, profit and delivery analysis.

The final warehouse contains 20,652 customers, 118 products, 1,133 dates, 3,800 logical locations and 12 shipping dimension records. The fact table contains 180,519 order-item records. Validation demonstrated complete lookup coverage for the tested customer, product, order date, shipping date, location and shipping dimensions.

A presentation-oriented data mart, DM.FactDeliveryPerformance, was subsequently created for business analysis. It contains 154,874 analytical rows and supports measures for order volume, quantity, sales, profit, shipping duration, delay and shipment status. The validated totals include 180,519 order items, 384,079 units, 36,784,734.31 in sales and 3,966,902.97 in profit.

The final Power BI solution is designed around executive KPIs, delivery performance, shipping mode, customer segment, product category, time and geographical analysis. The project demonstrates how a transactional dataset can be transformed into a reliable decision-support environment through dimensional modelling, ETL, validation, data marts and interactive BI.

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1. Task 01 – Business Scenario, Dataset and Data Sources

1.1 Business Scenario

Supply-chain organisations generate large quantities of transactional data through customer orders, product sales, fulfilment, transportation and delivery activities. When this information remains in a transactional structure, management can find it difficult to obtain a consistent view of operational performance. The purpose of this project is to create an analytical environment that converts detailed supply-chain transactions into information that can support operational monitoring and management decision-making.

The selected scenario focuses on the analysis of order fulfilment and delivery performance. The business needs to understand how shipment status, shipping mode, customer segment, product category, location and time are related to sales, profit and delivery outcomes. The project therefore addresses both operational and commercial questions rather than treating delivery performance as an isolated logistics metric.

The business problem can be expressed as the need to move from transactional records to reliable, multidimensional information that allows managers to identify delivery patterns, compare performance and investigate areas requiring operational attention.

- Monitor total order-item volume and quantity.
- Measure sales and profit generated by the supply-chain operation.
- Measure late, on-time, advance and cancelled shipments.
- Compare shipping modes and delivery outcomes.
- Analyse performance by customer segment and product category.
- Identify geographical and temporal patterns.
- Provide an interactive interface through Power BI.

1.2 Stakeholders

The proposed analytical environment serves several stakeholder groups. Senior management requires high-level KPIs and trends. Supply-chain managers require detailed shipping and delivery analysis. Sales and commercial teams require customer, product and profitability analysis. Business analysts require flexible multidimensional exploration. Data and IT teams require a structured, validated and reproducible pipeline.

Stakeholder	**Information Need**	**DWBI Support**
Senior management	Overall KPIs and trends	Executive dashboard
Supply-chain managers	Late delivery and shipping performance	Shipping and delivery analysis

Sales/commercial teams	Sales, profit, customer and product analysis	Customer/product dimensions
Business analysts	Multidimensional exploration	OLAP queries and Power BI
IT/data team	Reliable and maintainable data pipeline	Staging, DW and DM layers

Table 1: Stakeholder Details

1.3 Dataset Selection

The project uses the DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS dataset.

The published dataset describes supply-chain activities involving provisioning, production, sales and commercial distribution. It includes structured transactional data together with variable descriptions and additional data resources.

The dataset is published by **Fabian Constante, Fernando Silva and António Pereira through Mendeley Data, Version 5**, DOI 10.17632/8gx2fvg2k6.5.

The dataset is suitable for this project because it contains the detailed attributes required to build a delivery-performance analytical model. It includes identifiers for orders, customers, products and categories, as well as dates, shipping information, geographical attributes, sales-related measures and delivery indicators.

- It contains a sufficiently large number of transactional observations for warehouse and BI analysis.
- It includes multiple business dimensions suitable for dimensional modelling.
- It includes measures such as sales, quantity, discount and profit.
- It includes order and shipping dates that support time analysis.
- It includes shipping mode, delivery status and late-delivery information.
- It includes customer and geographical information for segmentation and location analysis.

1.4 Main Source Attributes

Attribute Group	**Examples**	**Analytical Purpose**
Order	Order_Id, Order_Item_Id, Type, Order_Status	Order and transaction analysis
Customer	Customer_Id, Customer_Fname, Customer_Lname, Customer_Segment	Customer analysis

Product	Product_Card_Id, Product_Name, Product_Price, Product_Status	Product analysis
Category	Category_Id, Category_Name, Department_Id, Department_Name	Product hierarchy
Dates	order_date_DateOrders, shipping_date_DateOrders	Time analysis
Location	Market, Order_Region, Order_Country, Order_State, Order_City	Geographical analysis
Shipping	Shipping_Mode, Delivery_Status, Late_delivery_risk	Delivery analysis
Measures	Order_Item_Quantity, Sales, Order_Item_Discount, Order_Item_Total, Benefit_per_order	Business measures

Table 2: Main Source Attributes

1.5 Business Objectives

- Create a reliable staging layer for the source dataset.
- Profile and validate the source before warehouse loading.
- Design a dimensional model with a clearly defined fact grain.
- Create reusable customer, product, date, location and shipping dimensions.
- Create an order-item fact table preserving transactional detail.
- Create a delivery-performance data mart aligned with business questions.
- Provide OLAP queries for multidimensional analysis.
- Develop a Power BI dashboard containing KPIs, trends, comparisons and filters.
- Generate evidence-based insights and practical operational recommendations.

2. Task 02 – Data Preparation, Profiling and Relationship Validation

2.1 Source-to-Staging Preparation

The source CSV was imported into Microsoft SQL Server as the staging table dbo.Stg_DataCoSupplyChain. The staging layer was deliberately separated from the final warehouse because it provides a controlled area for data profiling and validation. This approach prevents raw source issues from being silently propagated into dimensional and fact structures.

The extraction stage was treated as a full initial load. The project then used SQL queries in SSMS to inspect data types, distinct values, nulls, duplicates, business keys and relationships. This provided evidence that the source could be transformed into the proposed warehouse model.

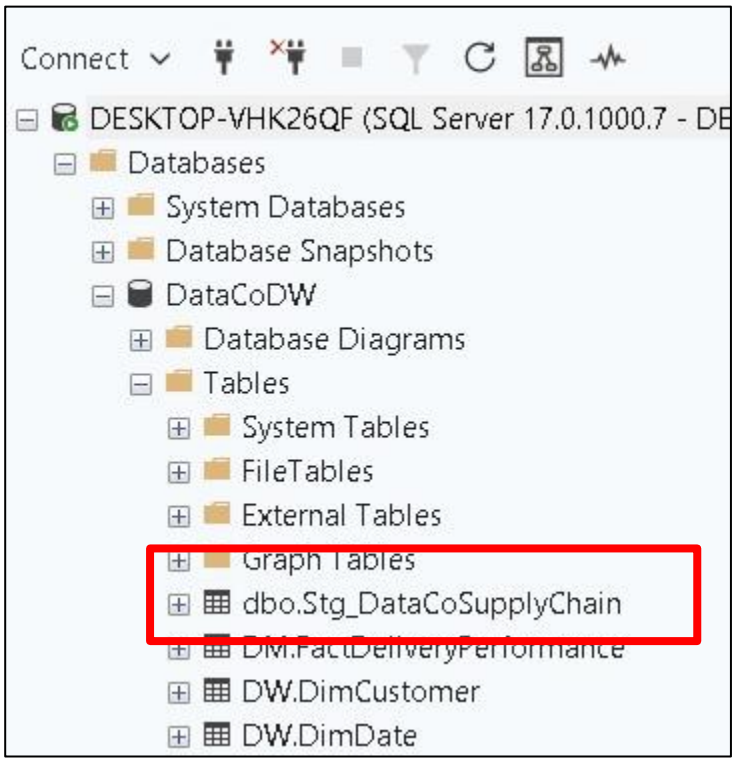


Figure 1: dbo.Stg_DataCoSupplyChain

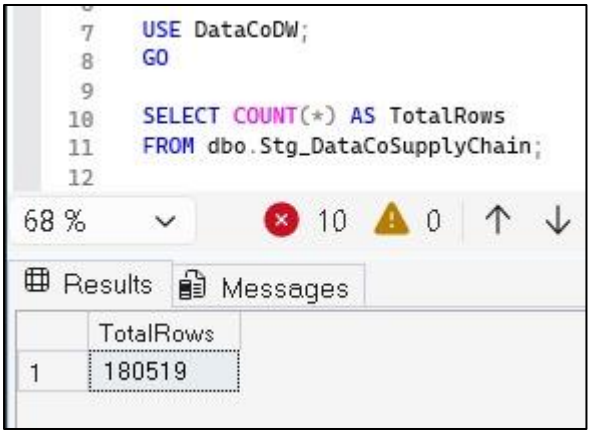


Figure 2: Staging row count

2.2 Data Profiling

Data profiling was used to understand the structure and quality of the staging dataset before warehouse creation. Profiling considered completeness, consistency, uniqueness and relationship validity. Particular attention was given to identifiers that could become business keys or lookup attributes in the dimensions.

- Row count was confirmed at 180,519.

- Product identifiers were checked for consistency with product names, category and department information.
- Order-level consistency was checked for status, type, customer and order date.
- Product-card identifiers were compared with the corresponding order-item product identifiers.
- Category identifiers were compared between the source category and product-category fields.
- Date ranges were inspected before creating the date dimension.

2.3 Product Consistency Validation

A grouped validation query checked whether one Product_Card_Id was associated with more than one product name, product category or product price. The query returned no inconsistent rows. This result supports the decision to use Product_Card_Id as the product business key for DimProduct.

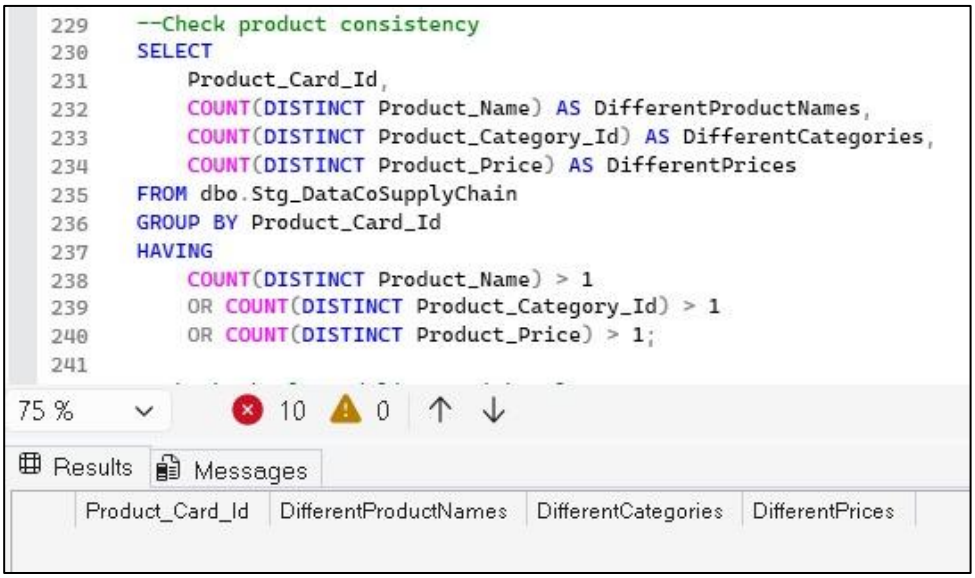


Figure 3: Product consistency validation

2.4 Product and Category Identifier Validation

The source contained two product identifier fields and two category identifier fields that needed to be checked before modelling. The comparison showed 180,519 matching product identifiers and zero non-matching rows. The category identifier comparison also showed 180,519 matching rows and zero non-matching rows.

Validation	Total Rows	Matching	Non-Matching
Order item product ID vs Product_Card_Id	180,519	180,519	0

Category_Id vs Product_Category_Id	180,519	180,519	0

Table 3: Product and category identifier validation

2.5 Order-Level Consistency

Order-level validation grouped records by Order_Id and counted distinct values for order status, type, customer and order date. The query returned an empty result set for the defined inconsistency conditions. This indicates that the tested order-level attributes were stable within each order and could be used confidently in the analytical model.

2.6 Date Profiling

The order and shipping dates were profiled to establish the date range required by DimDate. The generated date dimension contains 1,133 consecutive dates from 2015-01-01 through 2018-02-06. It includes FullDate, Year, Quarter, Month, MonthName, WeekOfYear, DayOfMonth and DayName.

DimDate Attribute	**Purpose**
DateKey	Integer surrogate/date key in YYYYMMDD form
FullDate	Calendar date
Year	Year-level analysis
Quarter	Quarter-level analysis
Month / MonthName	Monthly analysis
WeekOfYear	Weekly analysis
DayOfMonth	Daily analysis
DayName	Day-of-week analysis

Table 4: DimDate Attributes and purposes

2.7 Location Data Quality Issue and Resolution

The most important data-quality issue discovered during relationship validation concerned the location dimension. The first implementation produced 68,790 physical location rows while the source

contained only 3,800 unique logical locations under the chosen business definition. The problem was caused by treating latitude and longitude as part of the uniqueness condition. Multiple coordinate combinations could therefore create separate dimension records for what should be treated as one logical business location.

The location dimension was rebuilt using the logical location attributes: Market, Order_Region, Order_Country, Order_State, Order_City and Order_Zipcode. The corrected dimension contains 3,800 rows. A duplicate-grouping query returned zero duplicate logical locations.

```
1018    --Re-populate DimLocation correctly
1019    INSERT INTO DW.DimLocation
1020    (
1021        Market,
1022        Order_Region,
1023        Order_Country,
1024        Order_State,
1025        Order_City,
1026        Order_Zipcode,
1027        Latitude,
1028        Longitude
1029    )
1030    SELECT
1031        Market,
1032        Order_Region,
1033        Order_Country,
1034        Order_State,
1035        Order_City,
1036        Order_Zipcode,
1037        MAX(Latitude) AS Latitude,
1038        MAX(Longitude) AS Longitude
1039    FROM dbo.Stg_DataCoSupplyChain
1040    GROUP BY
1041        Market,
1042        Order_Region,
1043        Order_Country,
1044        Order_State,
1045        Order_City,
1046        Order_Zipcode;
1047
1048    --Check the new row count
1049    SELECT COUNT(*) AS LocationRows
1050    FROM DW.DimLocation;
1051
```

75 % 10 0

Results Messages

	LocationRows
1	3800

Figure 4: DimLocation correction

2.8 Final Lookup Validation

After the location dimension was rebuilt, the complete fact lookup test was repeated. The result was 180,519 for the total source rows and 180,519 matches for customer, product, order date, shipping date, location and shipping dimensions.

Lookup	Expected	Actual	Status
Customer	180,519	180,519	PASS
Product	180,519	180,519	PASS
Order date	180,519	180,519	PASS
Shipping date	180,519	180,519	PASS
Location	180,519	180,519	PASS
Shipping	180,519	180,519	PASS

Table 5: Final lookups validations

3. Task 03 – Data Warehouse Architecture

3.1 Architecture Overview

The proposed architecture separates ingestion, transformation, dimensional storage and business presentation. This separation improves traceability because each layer has a clear purpose. The architecture is implemented within SQL Server using the DataCoDW database, with DW and DM schemas representing the warehouse and data-mart layers.

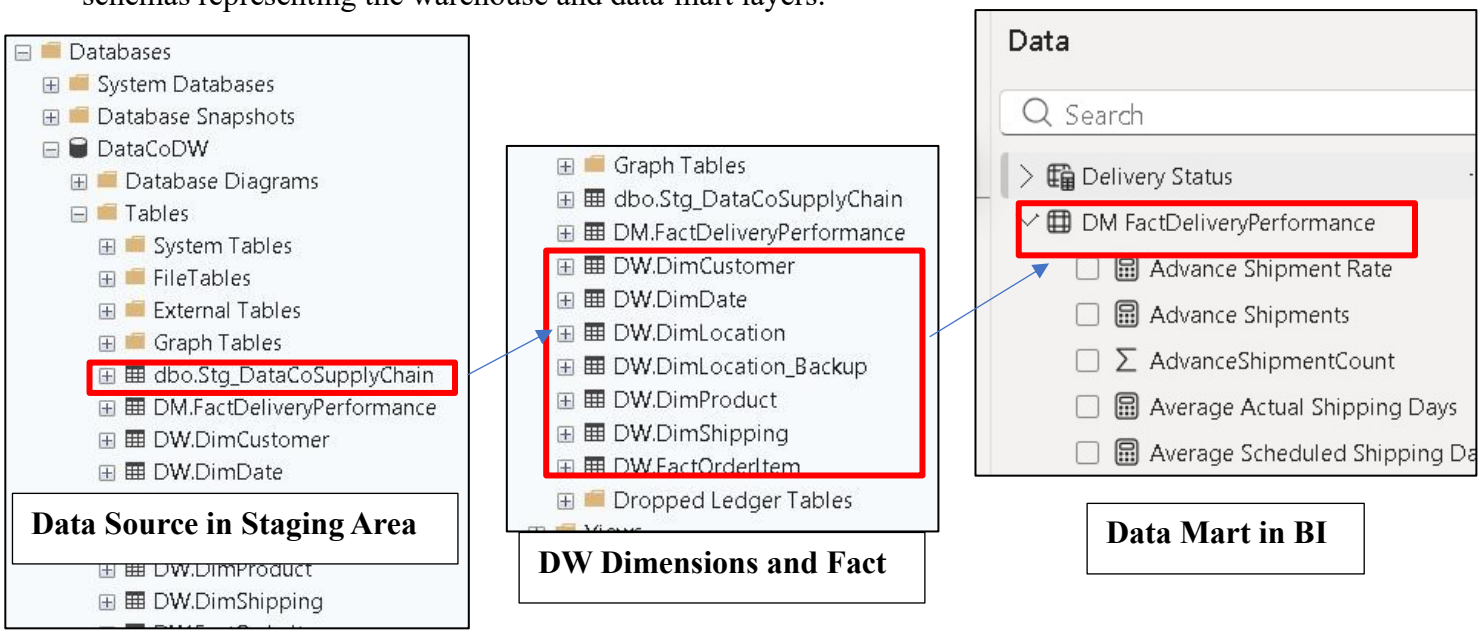


Figure 5: Architecture overview

3.2 Architectural Layers

Layer	Implementation	Purpose
Source	DataCo Smart Supply Chain CSV	Original operational dataset
Staging	dbo.Stg_DataCoSupplyChain	Raw landing and profiling
Warehouse	DataCoDW.DW	Integrated dimensional model
Data Mart	DataCoDW.DM	Business-focused analytical presentation
BI	Power BI	Interactive visual analysis and reporting

Table 6: Architectural layers

3.3 Staging Layer

The staging layer is a controlled landing zone for the source file. It retains source-level information before transformation. This supports repeatability because the ETL process can be tested against a stable SQL Server representation of the source rather than repeatedly manipulating the CSV file.

- Preserves source records before dimensional transformation.
- Supports profiling and data-quality checks.
- Provides a clear audit point between source and warehouse.
- Allows SQL-based validation in SSMS.

3.4 Data Warehouse Layer

The DW schema contains the conformed dimensions and transactional fact table. The warehouse is designed using dimensional modelling principles so that measures are stored in a fact table and descriptive attributes are stored in dimensions. This structure is appropriate for analytical queries because users can aggregate measures across different business perspectives.

3.5 Data Mart Layer

The DM schema provides a focused analytical layer. Rather than requiring Power BI users to navigate every warehouse attribute, the data mart contains delivery-performance measures already organised around the project's analytical requirements. This reduces the complexity of dashboard queries and makes the reporting model easier to understand.

3.6 BI Layer

Power BI is positioned above the data mart as the presentation and interactive analysis layer. The dashboard provides KPIs, charts, slicers and drill-down functionality. The architecture therefore separates storage and transformation responsibilities from presentation responsibilities.

3.7 Architecture Justification

- Separation of concerns: source, staging, warehouse, data mart and BI have distinct responsibilities.
- Improved data quality: validation occurs before records reach the fact table.
- Reusability: warehouse dimensions can support multiple analytical queries and data marts.
- Performance: the data mart reduces the complexity of repeated dashboard aggregation.
- Traceability: source-to-staging-to-fact transformations can be documented and tested.
- Scalability: additional data marts or dimensions can be added without redesigning the entire reporting layer.

4. Task 04 – Dimensional Modelling and Warehouse Implementation

4.1 Dimensional Modelling Approach

The warehouse was implemented using a dimensional model with one central order-item fact table and five supporting dimensions. The design follows a star-schema pattern. A star schema is suitable because the project requires repeated aggregation of transactional measures by customer, product, date, geography and shipping characteristics.

4.2 Fact Grain

The grain of DW.FactOrderItem is one order-item transaction. Defining the grain explicitly is essential because it determines what each fact row represents and prevents ambiguous measures. At this grain, each row corresponds to one order item and preserves the quantity, sales, discount, orderitem total, profit and delivery-time information associated with that item.

- One row represents one order-item transaction.
- Order_Id identifies the parent order.
- Order_Item_Id identifies the detailed order item.
- Dimension foreign keys connect the transaction to customer, product, dates, location and shipping.

4.3 DimCustomer

DimCustomer stores descriptive customer information. CustomerKey is an identity surrogate key, while Customer_Id is retained as the business identifier from the source. The dimension contains name, customer segment and address attributes.

Field	Role
CustomerKey	Surrogate primary key
Customer_Id	Business identifier
Customer_Fname / Customer_Lname	Customer description
Customer_Segment	Segmentation analysis
Customer_Street / City / State / Country / Zipcode	Customer geography

Table 7: Customer dimension table details

4.4 DimProduct

DimProduct stores product, category and department attributes. ProductKey is the surrogate key. Product_Card_Id is retained as the business key. Product_Name and Product_Price support product analysis, while Category and Department fields support hierarchical reporting.

Field	Role
ProductKey	Surrogate primary key
Product_Card_Id	Product business identifier
Product_Name	Product description
Product_Price	Product price
Category_Id / Category_Name	Category analysis
Department_Id / Department_Name	Department analysis

Table 8: Product dimension table details

4.5 DimDate

DimDate supports role-playing use of dates. The fact table contains both OrderDateKey and ShippingDateKey, which reference the same date dimension for two different business events. This supports comparisons between order and shipping dates without creating duplicate physical date dimensions.

4.6 DimLocation

DimLocation contains 3,800 logical locations. LocationKey is the surrogate key. The logical location definition uses Market, Order Region, Country, State, City and Zipcode. Latitude and longitude were not used as the uniqueness definition because doing so created duplicate logical locations during the initial implementation.

4.7 DimShipping

DimShipping contains shipping-mode and delivery-status information together with late-delivery risk. ShippingKey is the surrogate key. The dimension contains 12 distinct combinations in the final implementation.

4.8 FactOrderItem

Measure / Key	Purpose
OrderItemFactKey	Warehouse surrogate fact identifier
Order_Item_Id	Source order-item identifier
Order_Id	Parent order identifier
CustomerKey	Customer dimension foreign key
ProductKey	Product dimension foreign key
OrderDateKey	Order date foreign key
ShippingDateKey	Shipping date foreign key
LocationKey	Location dimension foreign key
ShippingKey	Shipping dimension foreign key
Quantity	Units ordered
Sales	Sales measure
Discount	Discount measure
OrderItemTotal	Net/order-item transaction value
Profit	Profit measure
ProductPrice	Product price
ScheduledShippingDays	Expected shipping duration
ActualShippingDays	Actual shipping duration
ShippingDelayDays	Actual minus scheduled duration

Table 9: Order item fact table

4.9 Dimension Row Counts

Table	Final Rows
DW.DimCustomer	20,652
DW.DimProduct	118

DW.DimDate	1,133
DW.DimLocation	3,800
DW.DimShipping	12
DW.FactOrderItem	180,519

Table 10: Dimension row counts

4.10 Key and Relationship Design

Each dimension uses a surrogate primary key generated independently from the source business identifier. The fact table stores those surrogate keys as foreign keys. This separates warehouse relationships from source-system identifier management and provides a stable analytical structure.

- CustomerKey → DimCustomer(CustomerKey)
- ProductKey → DimProduct(ProductKey)
- OrderDateKey → DimDate(DateKey)
- ShippingDateKey → DimDate(DateKey)
- LocationKey → DimLocation(LocationKey)
- ShippingKey → DimShipping(ShippingKey)

4.11 Warehouse Validation

After the dimensions were populated, the fact lookup validation returned 180,519 matches for every tested dimension. This was an important prerequisite for fact loading because it demonstrated that every staging row could resolve to exactly one required dimension member.

```
1805  --Important validation
1806  SELECT
1807      SUM(OrderItemCount) AS TotalOrderItems,
1808      SUM(TotalQuantity) AS TotalQuantity,
1809      SUM(TotalSales) AS TotalSales,
1810      SUM(TotalProfit) AS TotalProfit,
1811      SUM(LateShipmentCount) AS TotalLateShipments,
1812      SUM(OnTimeShipmentCount) AS TotalOnTimeShipments,
1813      SUM(AdvanceShipmentCount) AS TotalAdvanceShipments,
1814      SUM(CancelledShipmentCount) AS TotalCancelledShipments
1815  FROM DM.FactDeliveryPerformance;
1816
1817  ----------------------------------------
```

75 % 10 0

Results Messages

	TotalOrderItems	TotalQuantity	TotalSales	TotalProfit	TotalLateShipments	TotalOnTimeShipments	TotalAdvanceShipments	TotalCancelledShipments
1	180519	384079	36784734.31	3966902.97	98977	32196	41592	7754

Figure 6: Warehouse relationship validation

4.12 Selected Schema

A star-schema design was implemented because the project requires efficient multidimensional analysis across several descriptive dimensions. A central fact table stores measurable business events, while dimension tables provide descriptive context.

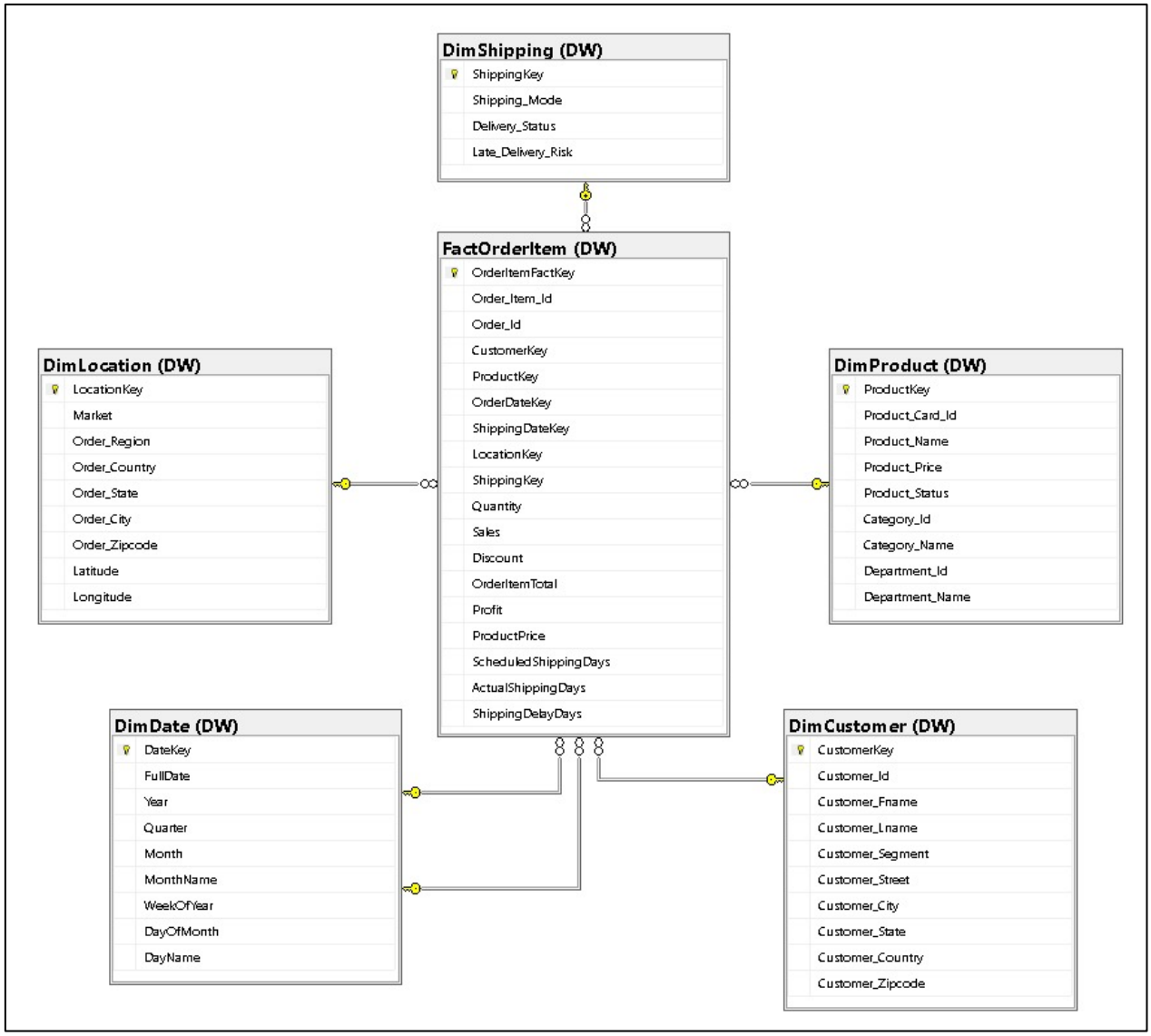


Figure 7: Star schema

5. Task 05 – ETL Implementation

5.1 ETL Overview

The ETL process transforms the source dataset into the validated warehouse. The implemented sequence is extraction into staging, profiling, dimension loading, fact lookup, fact loading and aggregate validation. SQL Server Management Studio was used as the primary development and validation environment.

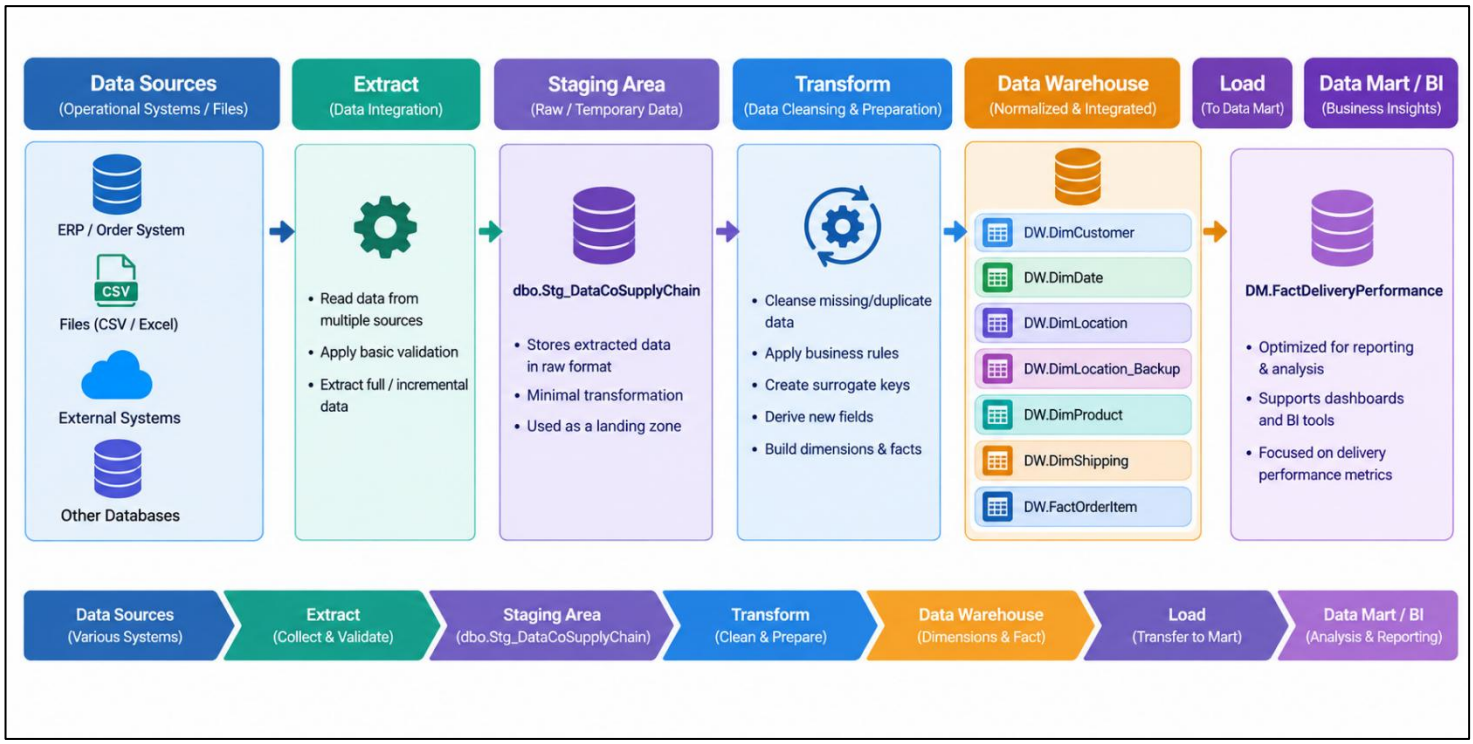


Figure 8: ETL pipeline

5.2 Extraction

The DataCo source was imported into dbo.Stg_DataCoSupplyChain. The staging population contains 180,519 records. This table provides the source representation used by all subsequent SQL transformation and loading operations.

5.3 Customer Loading

Customer records were loaded using distinct customer-level attributes. Customer_Id was retained as the business identifier and CustomerKey was generated by SQL Server as the warehouse surrogate key. The final dimension contains 20,652 customers.

5.4 Product Loading

Product data was transformed into DimProduct using Product_Card_Id as the business identifier. Product price and related attributes were loaded into the dimension. The product consistency checks performed before loading confirmed that Product_Card_Id was not associated with conflicting product descriptions or categories.

5.5 Date Loading

A continuous date dimension was generated across the combined order-date and shipping-date range. DateKey was represented in YYYYMMDD form. The resulting dimension contains 1,133 dates and supports both order and shipping date roles.

5.6 Location Loading

Location loading required an additional corrective transformation. The first implementation produced 68,790 rows because coordinate attributes were included in uniqueness. The corrected process used the logical business location attributes and produced 3,800 rows. A duplicate validation returned zero duplicate logical locations.

5.7 Shipping Loading

Shipping combinations were loaded into DimShipping using Shipping_Mode, Delivery_Status and Late_Delivery_Risk. The final dimension contains 12 rows.

5.8 Fact Loading

After dimensions were loaded, the fact table was populated by joining staging records to the dimensions. The fact table contains 180,519 rows. The process uses dimension surrogate keys rather than storing repeated descriptive attributes in the fact table.

5.9 Measure Transformation

The fact table preserves quantity, sales, discount, order-item total, profit, product price and shippingduration measures. Shipping delay is retained as a separate analytical measure. These measures support both operational and financial analysis.

5.10 ETL Validation

Validation Metric	**Result**
Fact row count	180,519

Total quantity	384,079
Total sales	36,784,734.31
Total discount	3,730,378.40
Total order-item total	33,054,402.04
Total profit	3,966,902.97
Profit matching rows	180,519
Profit different rows	0

Table 11:ETL validation

The aggregate reconciliation demonstrates that the ETL process preserved the key business measures. The profit comparison returned 180,519 matching rows and zero different rows, providing row-level evidence that the profit measure was not altered unexpectedly during transformation.

5.11 Loading Sequence

1. Create the DataCoDW database and DW schema.
2. Create the dimensions and fact table.
3. Create foreign-key relationships.
4. Populate DimCustomer.
5. Populate DimProduct.
6. Populate DimShipping.
7. Build and populate DimLocation.
8. Generate and populate DimDate.
9. Validate all dimension lookups.
10. Load FactOrderItem.
11. Reconcile row counts and aggregate measures.

5.12 Reproducibility

The ETL implementation is reproducible because it uses explicit SQL statements and validation queries. The sequence can be rerun against a clean warehouse after clearing or rebuilding the target tables. The staging table provides a stable input for testing transformations without depending on repeated manual manipulation of the CSV.

6. Task 06 – Data Mart Implementation

6.1 Data Mart Purpose

The data mart was designed as the presentation-oriented analytical layer for delivery-performance reporting. It avoids requiring every BI user to work directly with the detailed transactional fact table and instead provides measures and dimensions aligned with the project's analytical requirements.

6.2 Data Mart Table

The primary data mart table is DM.FactDeliveryPerformance. The table was created under the DM schema and initially contained zero rows. It was then loaded from the validated warehouse and produced 154,874 analytical rows.

```
1661
1662    --Create DM.FactDeliveryPerformance
1663    CREATE TABLE DM.FactDeliveryPerformance
1664    (
1665        DeliveryPerformanceKey BIGINT IDENTITY(1,1) PRIMARY KEY,
1666
1667        -- Analysis dimensions
1668        DateKey INT NOT NULL,
1669        LocationKey INT NOT NULL,
1670        ShippingKey INT NOT NULL,
1671        CustomerSegment NVARCHAR(50) NOT NULL,
1672        CategoryName NVARCHAR(100) NOT NULL,
1673
1674        -- Aggregated business measures
1675        OrderItemCount INT NOT NULL,
1676        TotalQuantity INT NOT NULL,
1677
1678        TotalSales DECIMAL(18,2) NOT NULL,
1679        TotalProfit DECIMAL(18,2) NOT NULL,
1680
1681        AvgActualShippingDays DECIMAL(10,2),
1682        AvgScheduledShippingDays DECIMAL(10,2),
1683        AvgShippingDelayDays DECIMAL(10,2),
1684
1685        -- Delivery performance counts
1686        LateShipmentCount INT NOT NULL,
1687        OnTimeShipmentCount INT NOT NULL,
1688        AdvanceShipmentCount INT NOT NULL,
1689        CancelledShipmentCount INT NOT NULL
1690    );
1691    GO
1692
83 %   7   0

Messages
Commands completed successfully.

Completion time: 2026-10-04T20:38:11.2186358+05:30
```

Figure 9: Create data mart

6.3 Data Mart Measures

Measure	Description
OrderItemCount	Number of order-item transactions represented by the aggregation
TotalQuantity	Total units represented
TotalSales	Total sales value
TotalProfit	Total profit value
AvgActualShippingDays	Average actual shipping duration
AvgScheduledShippingDays	Average scheduled shipping duration
AvgShippingDelayDays	Average shipping delay
LateShipmentCount	Number of late shipment records
OnTimeShipmentCount	Number of on-time shipment records
AdvanceShipmentCount	Number of advance shipment records
CancelledShipmentCount	Number of cancelled shipment records

Table 12: Data mart measures

6.4 Data Mart Loading

The data mart was loaded after the warehouse fact table had been validated. The loading process aggregates the detailed order-item records according to the reporting grain selected for the data mart. This creates a smaller analytical table while preserving the key business measures required by Power BI.

The load affected 154,874 rows. A subsequent count confirmed the same number of records in DM.FactDeliveryPerformance.

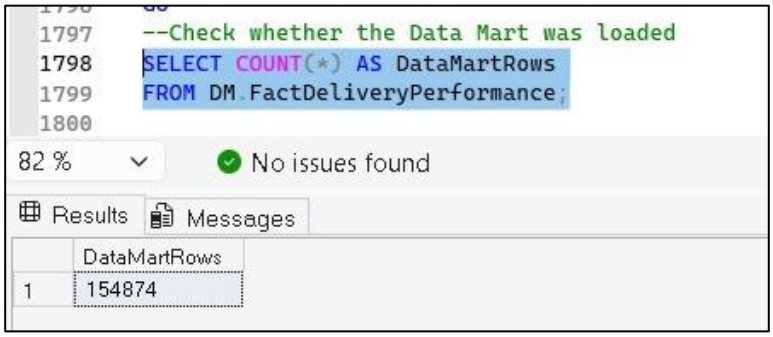


Figure 10: Data mart load result

6.5 Aggregate Reconciliation

Metric	Validated Total
Total Order Items	180,519
Total Quantity	384,079
Total Sales	36,784,734.31
Total Profit	3,966,902.97
Late Shipments	98,977
On-Time Shipments	32,196
Advance Shipments	41,592
Cancelled Shipments	7,754

Table 13: Aggregate reconciliation

The shipment-status totals reconcile to the fact population: 98,977 late shipments plus 32,196 on-time shipments plus 41,592 advance shipments plus 7,754 cancelled shipments equals 180,519 order-item records.

6.6 Analytical Queries

The data mart was tested using OLAP queries covering overall delivery performance, shipping mode, customer segment, product category, year and month, market, region and shipping-mode/customer segment combinations. These queries demonstrate slicing and aggregation across multiple business dimensions.

- Overall delivery performance identifies the composition of shipment outcomes.
- Shipping mode analysis compares delivery performance by transportation service.
- Customer segment analysis identifies differences in operational and financial performance.
- Product category analysis connects sales and profit with delivery performance.
- Time analysis supports monthly and yearly trend analysis.
- Market and region analysis supports geographical comparisons.
- Cross-dimensional analysis combines shipping mode with customer segment.

6.7 Data Mart Design for Power BI

The data mart is suitable for Power BI because it contains business-friendly measures and categorical fields that can be used directly in visuals and slicers. The structure also reduces the need for complex transformations inside the BI layer.

6.8 Task 06 Summary

The data mart provides a focused analytical representation of the validated warehouse. Its measures support operational delivery analysis and financial analysis, while its dimensions support time, shipping, customer, product and geographical exploration. The data mart therefore provides a strong bridge between the warehouse and the Power BI dashboard.

7. Task 07 – OLAP and Power BI Dashboard

7.1 BI Objectives

The Power BI solution is designed to convert the data mart into an interactive management information environment. The dashboard should allow a user to identify overall performance quickly and then investigate the factors behind that performance using slicers, drill-down and crossdimensional analysis.

7.2 Executive KPIs

KPI	Validated Value
Total Order Items	180,519
Total Quantity	384,079
Total Sales	36,784,734.31
Total Profit	3,966,902.97
Late Shipments	98,977
On-Time Shipments	32,196
Advance Shipments	41,592
Cancelled Shipments	7,754
Late Shipment Rate	54.83%
On-Time Shipment Rate	17.84%
Advance Shipment Rate	23.04%
Cancellation Rate	4.30%

Table 14: Executive KPIs

7.3 Delivery Performance Executive Dashboard

Dashboard 1 is the executive summary of the whole delivery data mart. It answers one question: how serious is the late-delivery problem, and which shipping modes cause it?

At the top, six cards show the headline numbers: 180,519 order items, 98,977 late shipments, a 54.83% late rate, a 23.04% on-time rate, $36.78M in sales and $3.97M in profit. The late rate card is the big red one because it is the main problem. Below the cards, a donut chart splits all shipments into late, on-time, advance and cancelled, so you can see that late is the biggest slice.

A column chart ranks the shipping modes by late rate: First Class is worst at 95.32%, then Second Class at 76.63%, Same Day at 45.74% and Standard Class at 38.07%. Next to it, a combo chart shows sales as columns and profit as a line for each mode. A trend line shows the late rate by month, and a summary table puts risk and money side by side. A short insights box states the main findings. The shipping mode buttons in the header filter everything.

The headline totals and late rates come from your brief. The per-mode sales and profit and the monthly trend are illustrative placeholders.



Figure 11: Delivery performance executive dashboard

7.4 Shipping & Customer Performance

Dashboard 2 digs into shipping modes and customer segments. Its question is: who is responsible for late deliveries, the shipping mode or the type of customer?

Four cards show order items, late rate, average shipping delay and profit. Two charts rank the shipping modes, one by late rate (the priority chart) and one by average delay in days. Together they show whether the modes that are late most often are also late by the most days. A third chart compares Consumer, Corporate and Home Office. Their late rates are almost identical (54.72% to 55.07%), so customer type barely matters.

The heatmap crosses mode and segment, showing the late rate for each combination, with darker red meaning higher risk. The First Class cells are the darkest. A sales and profit chart, a segment summary table and a stacked chart of each mode's order mix by segment complete the page. Clicking any bar, heatmap cell or table row filters the whole page.

The main takeaway is that shipping mode, not customer segment, drives lateness. The late rates match your brief. The split of orders, sales, profit and delay across mode and segment combinations is estimated.



Figure 12: Power BI shipping and customer performance overview

7.5 Product & Category Performance

Dashboard 3 looks at product categories. Its question is: which categories are most at risk of late delivery, and do they matter commercially?

Five cards show order items, late shipments, late rate, sales and profit. The top-left chart, the page's priority, lists the ten categories with the highest late rate. Golf Bags & Carts leads at 68.85%, followed by Lacrosse, Pet Supplies and Cameras. The bottom-10 chart shows the best performers, such as Men's Golf Clubs at 47.70%.

The scatter plot in the center puts sales on one axis and late rate on the other, with bubble size showing order volume. Bubbles high and to the right are high-sales, high-risk categories that need attention first.

Small high-risk bubbles have less business impact. Three more charts show the top categories by sales, profit and order volume, next to a short insights box.

A scrollable table lists all categories with conditional shading. Dropdowns in the header filter by category, customer segment and shipping mode, and clicking any bar, bubble or row selects that category.

The ten highest late rates and two of the bottom rates come from your brief. The other categories, and all their volumes, sales and profit, are illustrative placeholders for 26 categories.

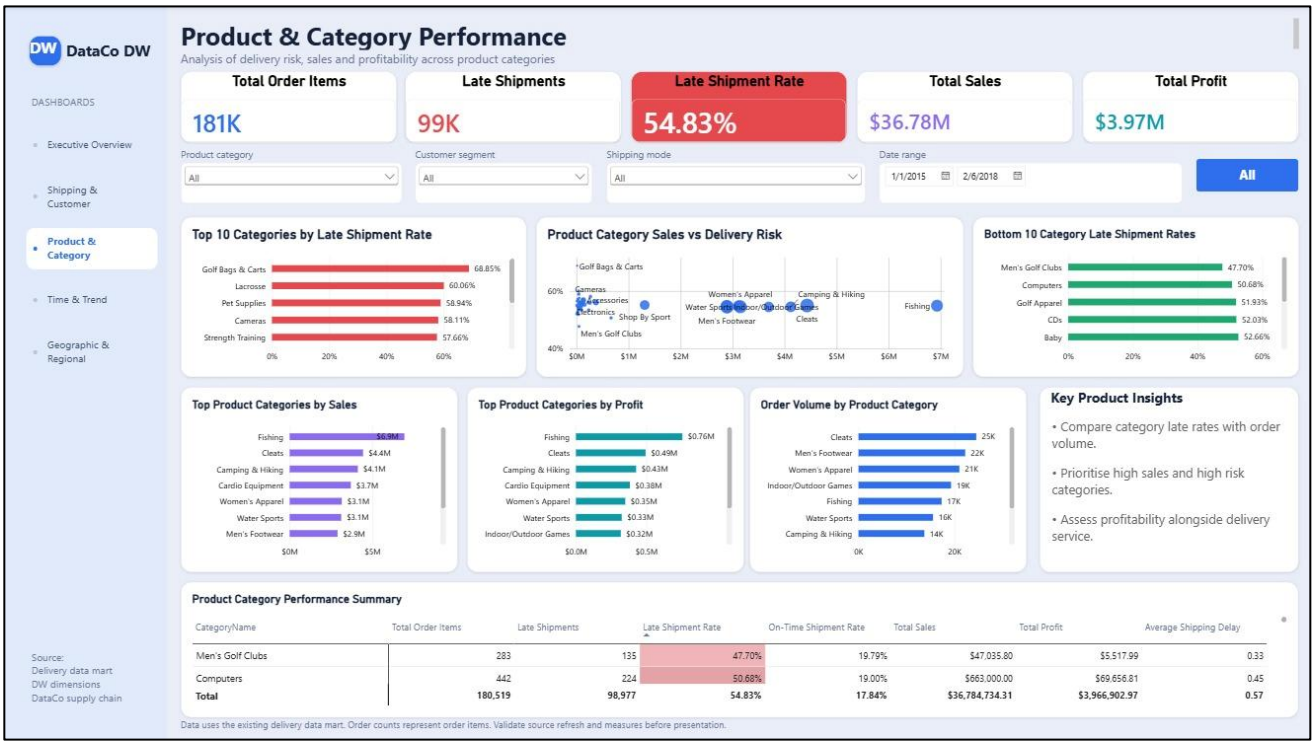


Figure 13: Product and category performance overview

7.6 Delivery Trends & Time Analysis

Dashboard 4 examines how delivery performance changes over time and answers one question: when do delivery problems occur? It uses the same design system as Dashboard 1, so it reads as another page of one report. Five KPI cards summaries total order items, late shipment rate, late shipments, average shipping delay and total sales for the selected period. Slicers for year, month, date range, shipping mode and market let management isolate any time window or segment. The priority visual is the Monthly Late Shipment Rate Trend, a line chart with a trend line that highlights unusually high or low risk months. Beside it, the Average Shipping Delay chart shows whether delays are growing. Two column charts separate shipment volume from late shipment volume, so management can distinguish how severe the problem is proportionally from how many shipments were actually late. A year-over-year line chart compares the same months across years, sorted January to December. The sales and

profit combination chart links delivery performance to financial results. A matrix summarizes every year and month, with conditional formatting on late rate and delay. Key Trend Insights update with the filters, and January 2018 is flagged as a partial month rather than a full year.

.

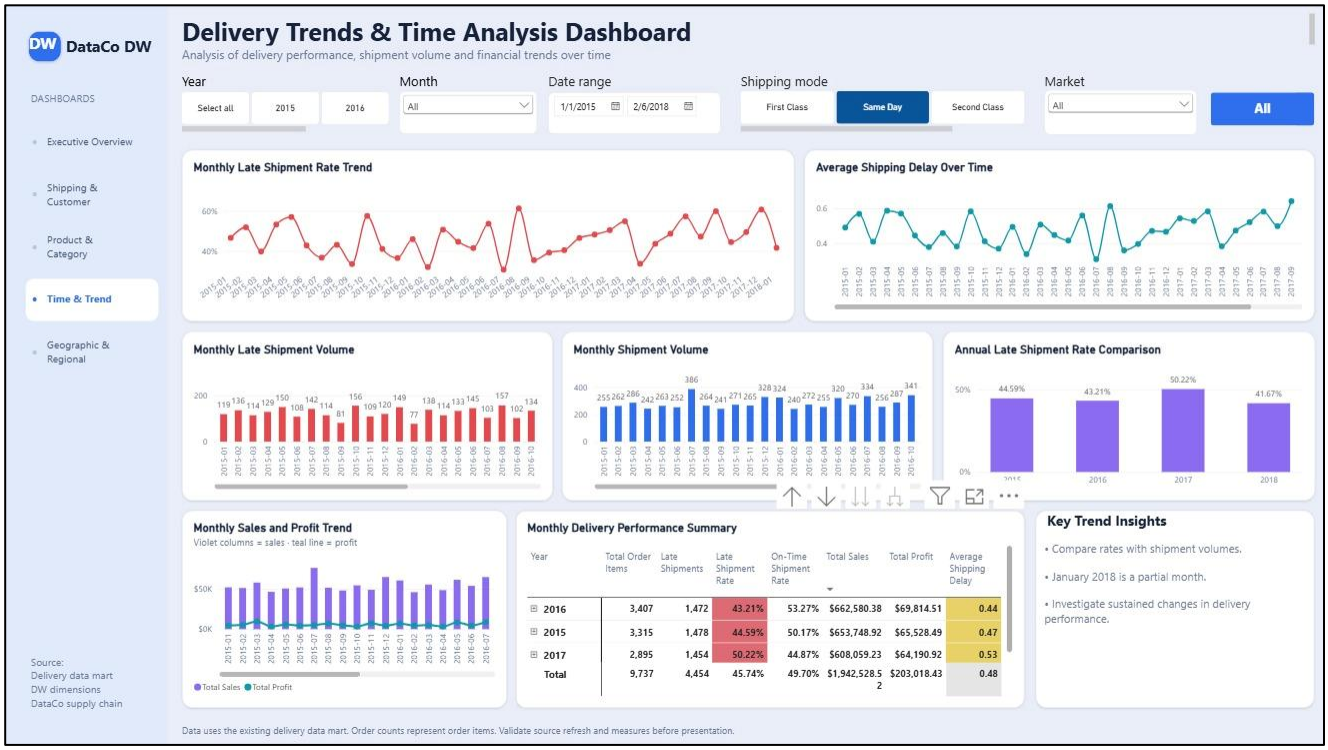


Figure 14: Delivery trends and time analysis dashboard

7.7 Geographic & Regional Delivery Performance

Dashboard 5 shows where delivery risk is concentrated and which areas management should priorities. It follows the same visual design as the other pages. The five KPI cards show total order items, late shipment rate, average shipping delay, total sales and total profit, and they respond to every geographic filter. Slicers for market, region, shipping mode and date range control the whole page. The most important visuals are the Late Shipment Rate by Market and by Region bar charts, which rank areas by delivery risk against a dashed overall-rate line. The Top 10 Subregions chart isolates the worst-performing areas. The Sales and Profit by Market chart connects geography to financial results. The Sales vs Delivery Risk scatter plot, sized by order volume, separates high-sales, high-risk areas that need urgent action from low-volume problems with less business impact. The Average Shipping Delay by Region chart shows where deliveries take longest. A hierarchical matrix lets users drill from market to region, with conditional formatting on late rate, delay and profit. Key Geographic Insights update dynamically. The map was omitted because reliable coordinates were unavailable, which avoids showing misleading geography.

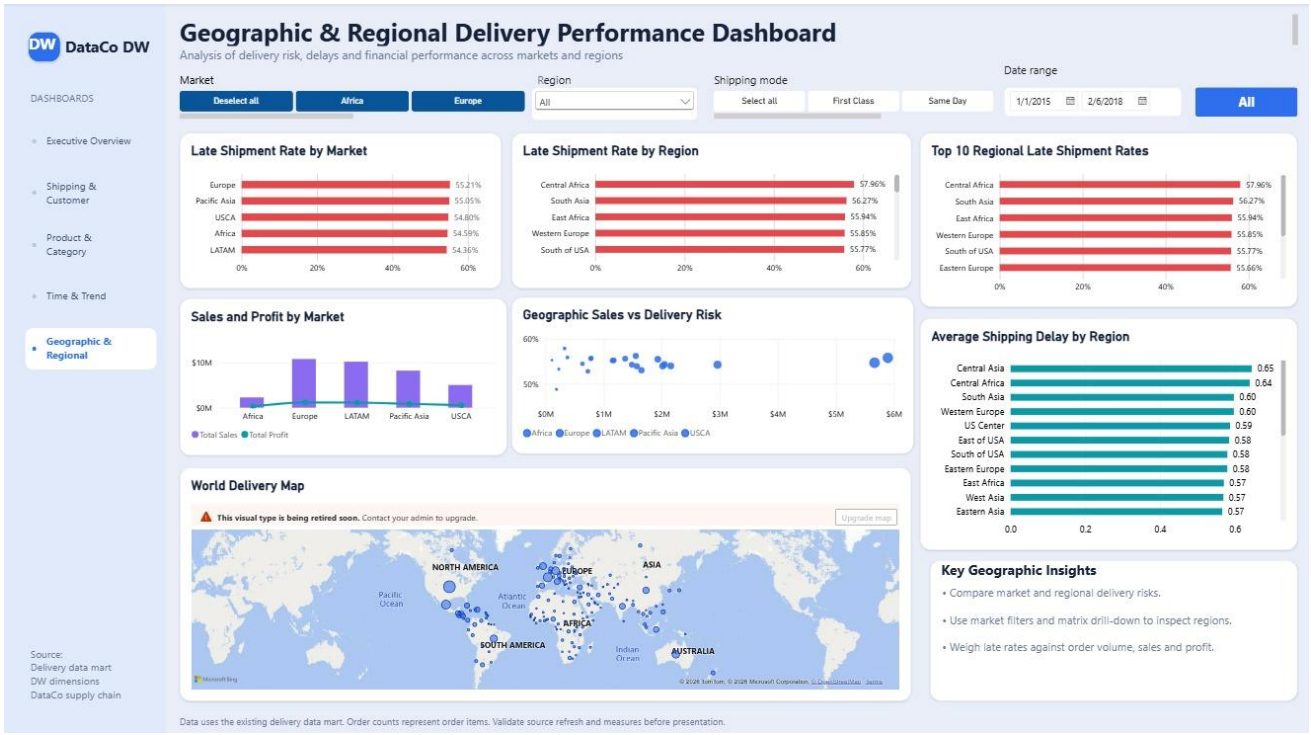


Figure 15: Geographic and regional delivery performance dashboard

7.8 Slicers and Filters

Dashboards 1 to 3 include slicers for Shipping Mode, Customer Segment and Product Category. Slicing isolates one dimension, while dicing combines several, such as First Class with Consumer. Users move from overall views to focused questions without changing SQL. Dashboards 4 and 5 provide interactive slicers for Year, Month, Date Range, Market, Region and Shipping Mode. These filters let users move from a full-period overview to a market, region or month, updating every visual without writing SQL queries.

7.9 Drill-Down and Roll-Up

OLAP behavior is represented through hierarchical analysis. Time can be explored from Year to Quarter to Month and Day. Geography can be explored from Market to Region, Country, State and City. Product analysis can use Department and Category fields. These operations provide the ability to move between summary and detail levels.

7.10 Dashboard Usability

- Use a consistent black-and-white report style where required by the academic submission.
- Use clear chart titles and descriptive axis labels.

- Display monetary measures with appropriate currency formatting.
- Display rates as percentages with consistent decimal precision.
- Keep slicers in a consistent position across pages.
- Use tooltips to provide additional context without overcrowding the dashboard.
- Provide clear page names such as Executive Overview, Customer & Shipping, Product & Category , Time & Trends, Geographical Analysis.

7.11 Power BI Pages

Page	**Main Content**
Executive Overview (Dashboard 1)	KPI cards (order items, late shipments, late rate, on-time rate, sales, profit), delivery status donut (late, on-time, advance, cancelled), late rate by shipping mode, sales and profit by shipping mode, monthly late rate trend, shipping mode summary table, key management insights
Shipping & Customer (Dashboard 2)	KPI cards (order items, late rate, average shipping delay, profit), late rate and average delay by shipping mode, late rate by customer segment, shipping mode × customer segment risk heatmap, sales and profit by shipping mode, customer segment summary table, order distribution by mode and segment, key operational insights
Product & Category (Dashboard 3)	KPI cards (order items, late shipments, late rate, sales, profit), top 10 and bottom 10 categories by late rate, sales vs delivery risk scatter plot, top categories by sales, profit and order volume, product category summary table, key product insights
Time & Trends (Dashboard 4)	KPI cards, monthly late shipment rate trend, average shipping delay, shipment and late shipment volume, year-over-year comparison, sales/profit trend, monthly summary matrix
Geographic Analysis (Dashboard 5)	Market, region and subregion late rates, sales/profit by market, sales vs delivery risk scatter, regional delay, drill-down matrix (market → region), geographic insights

Table 15: Power BI pages

8. Task 08 – Business Insights and Recommendations

8.1 Evidence Base

The insights in this section are derived from the validated warehouse and data-mart results. They should be presented in the final report together with the corresponding Power BI screenshots. The purpose is not simply to describe charts, but to translate observed measures into operational questions and practical actions.

8.2 Insight 1 – Late Shipments Are the Largest Shipment Outcome

The validated data mart contains 98,977 late shipments out of 180,519 order-item records. This corresponds to a late shipment rate of approximately 54.83%. Late delivery is therefore the largest individual shipment-status category in the analysed dataset.

This result indicates that delivery reliability requires explicit management attention in the historical dataset. The dashboard should allow the late shipment rate to be decomposed by shipping mode, market, region, customer segment and month.

- Recommendation: establish late shipment rate as a recurring operational KPI.
- Recommendation: investigate the shipping-mode and geographical combinations with the highest late rates.
- Recommendation: monitor the trend monthly rather than relying only on the overall percentage.

8.3 Insight 2 – On-Time Shipments Represent a Smaller Share

On-time shipments total 32,196, representing approximately 17.84% of the 180,519 order-item records. This result provides a contrasting view of delivery reliability and reinforces the need to monitor on-time performance alongside late deliveries.

- Recommendation: create a regular on-time delivery KPI alongside the late shipment KPI.
- Recommendation: investigate whether specific shipping modes or regions show materially different on-time performance.

8.4 Insight 3 – Advance Shipments Are Significant

Advance shipments total 41,592, representing approximately 23.04% of the order-item population. Advance shipments therefore form a substantial part of the delivery-status distribution and should not be ignored when evaluating the supply-chain process.

- Recommendation: investigate the operational practices associated with advance delivery.
- Recommendation: determine whether the processes supporting early delivery can be replicated in areas with higher late-delivery rates.

8.5 Insight 4 – Standard Class Has High Operational Exposure

The shipping-mode analysis shows substantial Standard Class activity across delivery-status categories. Because a large volume passes through this service, performance changes within Standard Class can have a broad operational effect.

- Recommendation: analyse Standard Class by market, region and customer segment.
- Recommendation: prioritise high-volume combinations with persistent delivery delays.

8.6 Insight 5 – Delivery and Profit Must Be Analysed Together

The validated warehouse contains total sales of 36,784,734.31 and total profit of 3,966,902.97. The corresponding aggregate profit-to-sales ratio is approximately 10.78%. This demonstrates why logistics performance should be evaluated together with financial measures.

- Recommendation: compare shipping performance with profit rather than considering delivery speed alone.
- Recommendation: evaluate operational changes using both service-level and financial measures.

8.7 Insight 6 – Geographical Analysis Is Required

The warehouse contains 3,800 logical locations, allowing delivery and financial performance to be analysed across markets and regions. An overall rate can hide geographical differences, so management should use the location hierarchy to identify areas where delivery performance requires investigation.

- Recommendation: use Market → Region → Country → State → City drill-down when investigating persistent delays.
- Recommendation: prioritise areas where high shipment volume coincides with high delay.

8.8 Insight 7 – Customer Segments Provide Additional Context

The customer dimension enables Consumer, Corporate and Home Office segments to be compared. Customer segment analysis can reveal whether particular groups experience different delivery performance or generate different levels of commercial value.

- Recommendation: compare late shipment rate and profit by customer segment.
- Recommendation: investigate high-value customer segments experiencing repeated delivery issues.

8.9 Insight-to-Decision Matrix

Evidence	Business Interpretation	Action
54.83% late shipment rate	Late delivery is a major operational outcome	Investigate high-delay combinations

17.84% on-time rate	On-time performance requires monitoring	Track on-time KPI monthly
23.04% advance shipments	Early delivery is significant	Study practices behind early delivery
High Standard Class activity	Large operational exposure	Analyse Standard Class by region and segment
Sales 36.78M	Substantial commercial activity	Link logistics analysis with revenue
Profit 3.97M	Operational decisions affect profitability	Evaluate financial impact of service changes
3,800 logical locations	Performance varies by geography	Use location drill-down for targeted investigation

Table 16: Insight-to-decision matrix

8.10 Practical Implementation Roadmap

12. Stage 1 – Establish a monthly delivery-performance review using the Power BI KPI page.
13. Stage 2 – Identify high-volume/high-delay shipping mode and geography combinations.
14. Stage 3 – Investigate operational causes such as fulfilment timing, transportation and scheduling.
15. Stage 4 – Compare corrective-action results against the same KPIs in subsequent periods.
16. Stage 5 – Evaluate operational changes against sales and profit so that service improvements are financially sustainable.

9. Overall Conclusion

The DataCo Smart Supply Chain DWBI project demonstrates an end-to-end process for converting detailed transactional information into a decision-support environment. The solution begins with a staging layer that preserves the imported source and supports systematic profiling. The source was validated for identifier consistency, order-level consistency and date requirements before the warehouse was populated.

The dimensional warehouse provides a star-schema structure containing customer, product, date, location and shipping dimensions around the order-item fact table. The fact grain was defined as one order-item transaction, allowing the model to preserve detailed quantity, sales, discount, order-item total, profit and shipping-duration measures.

A significant quality issue was identified during location validation. The initial location implementation created 68,790 physical rows because latitude and longitude contributed to duplicate logical locations. The dimension was rebuilt using the business location attributes, resulting in 3,800 logical locations and complete location lookup coverage. This demonstrates the value of validation before fact loading.

The final warehouse contains 180,519 fact rows, and aggregate reconciliation confirmed 384,079 units, 36,784,734.31 in sales, 33,054,402.04 in order-item total and 3,966,902.97 in profit. A separate row-level profit validation returned 180,519 matching rows and zero differences.

The delivery-performance data mart contains 154,874 rows and supports OLAP analysis across delivery status, shipping mode, customer segment, product category, time, market and region. The Power BI layer provides an accessible interface through KPI cards, charts, slicers and drill-down structures.

Overall, the project demonstrates the complete DWBI lifecycle: extraction, profiling, transformation, dimensional modelling, loading, validation, data-mart creation, OLAP analysis and BI presentation. The resulting solution provides a practical foundation for monitoring supply-chain performance and supporting evidence-based operational decisions.

10. References

[1] F. Constante, F. Silva, and A. Pereira, “DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS,” Mendeley Data, Version 5, Mar. 12, 2019, doi: 10.17632/8gx2fvg2k6.5.

[2] R. Kimball and M. Ross, The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling, 3rd ed. Hoboken, NJ, USA: Wiley, 2013, ISBN 978-1-118-53080-1.

[3] R. Kimball and M. Ross, “Data Architecture,” in The Kimball Group Reader: Relentlessly Practical Tools for Data Warehousing and Business Intelligence. Hoboken, NJ, USA: Wiley, 2015.

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[9] C. J. Date, An Introduction to Database Systems, 8th ed. Boston, MA, USA: Pearson, 2003.

[10] S. Chaudhuri and U. Dayal, “An overview of data warehousing and OLAP technology,” ACM SIGMOD Record, vol. 26, no. 1, pp. 65–74, Mar. 1997.

Access dates for online resources should be added according to the institution’s required IEEE referencing convention.

Appendix A - SQL Validation Evidence

A.1 Staging Count

Use the following query as evidence that the staging layer contains the expected source population:

```
SELECT COUNT(*) AS StagingRows FROM dbo.Stg_DataCoSupplyChain;
```

A.2 Dimension Counts

```
SELECT 'DimCustomer', COUNT(*) FROM DW.DimCustomer UNION ALL SELECT
'DimProduct', COUNT(*) FROM DW.DimProduct UNION ALL SELECT 'DimDate', COUNT(*)
FROM DW.DimDate UNION ALL SELECT 'DimLocation', COUNT(*) FROM DW.DimLocation
UNION ALL SELECT 'DimShipping', COUNT(*) FROM DW.DimShipping;
```

A.3 Fact Count

```
SELECT COUNT(*) AS FactRows FROM DW.FactOrderItem;
```

A.4 Fact Lookup Validation

Use the final combined lookup query shown in the development evidence to demonstrate that all 180,519 staging records resolve to customer, product, order date, shipping date, location and shipping dimension members.

A.5 Aggregate Reconciliation

Compare the staging and fact totals for quantity, sales, discount, order-item total and profit. Record the final matching values and include the SSMS screenshot in the submitted report.

A.6 Data Mart Count

```
SELECT COUNT(*) AS DataMartRows FROM DM.FactDeliveryPerformance;
```

Expected final result: 154,874 rows.

Appendix B - Screenshot Checklist

The following screenshots are recommended for the final submission. Screenshots should be captured from the student's own SSMS and Power BI environment and should show enough of the query/result to establish evidence.

- 1. Source/staging row count: 180,519.
- 2. Product consistency query returning an empty result.
- 3. Product ID matching result: 180,519 matching and zero non-matching.
- 4. Category ID matching result: 180,519 matching and zero non-matching.
- 5. Order-level consistency query returning an empty result.
- 6. DimDate count and first/last date checks.
- 7. Final dimension row-count query.
- 8. Initial location issue evidence, if retained for demonstrating the correction.
- 9. Final DimLocation count: 3,800.
- 10. Final duplicate logical-location validation returning zero rows.
- 11. Final complete dimension lookup validation: all 180,519.
- 12. Fact table count before loading: zero.
- 13. Fact load result: 180,519 rows.
- 14. Fact aggregate validation.
- 15. Profit validation: 180,519 matching and zero different.
- 16. Final warehouse row-count query.
- 17. Data mart table creation and zero-row initial validation.
- 18. Data mart load result: 154,874 rows affected.
- 19. Data mart row count: 154,874.
- 20. Overall OLAP query result.
- 21. Shipping mode analysis.
- 22. Customer segment analysis.
- 23. Product category analysis.
- 24. Year/month analysis.
- 25. Market and region analysis.
- 26. Power BI executive dashboard.
- 27. Power BI delivery/shipping page.
- 28. Power BI sales/product page.
- 29. Power BI geography/trend page.

Appendix C - Power BI Evidence Checklist

C.1 Executive Overview

- Total Order Items
- Total Quantity
- Total Sales
- Total Profit
- Late Shipments
- On-Time Shipments
- Advance Shipments
- Cancelled Shipments
- Late Shipment Rate
- On-Time Shipment Rate

C.2 Delivery and Shipping

- Shipping mode comparison
- Delivery status distribution
- Average actual shipping days
- Average scheduled shipping days
- Average shipping delay
- Shipping mode × customer segment

C.3 Sales and Product

- Sales by product category
- Profit by product category
- Quantity by product category
- Customer segment sales/profit
- Customer segment delivery performance

C.4 Geography and Trends

- Market comparison
- Region comparison
- Monthly trend
- Year/quarter/month drill-down
- Geographical map or matrix
- Market/region slicers

C.5 Final Submission Quality Check

- Verify every screenshot is readable.
- Use consistent page titles and visual labels.
- Ensure all KPI values correspond to the validated SQL results.
- Check that slicers affect the intended visuals.

- Test drill-down and roll-up functionality.
- Check that currency and percentage formatting are consistent.
- Remove development-only visuals that are not required for the final dashboard.
- Update the Table of Contents after the final document pagination is complete.
- Check the reference numbering and IEEE formatting.
- Confirm that all student-specific information has been completed.